

CEE 406 Pavement Design I
Department of Civil and Environmental Engineering
University of Illinois at Urbana-Champaign
Homework 4
Due: 10/08/2024

1. Solve Problem 2-6, Huang, pg. 92
2. Solve Problem 2-7, Huang, pg. 92
3. Plot the variation of vertical stress, vertical strain, horizontal stress, horizontal strain, and deflection with pavement depth (z) under the loading wheel center for the pavement structure below. Assume fully bonded layers. Please share your observations about the results (at least four observations). Please report the sign convention as well (i.e., indicate the meaning of positive values: compression or tension). You may use analysis software like winJULEA.

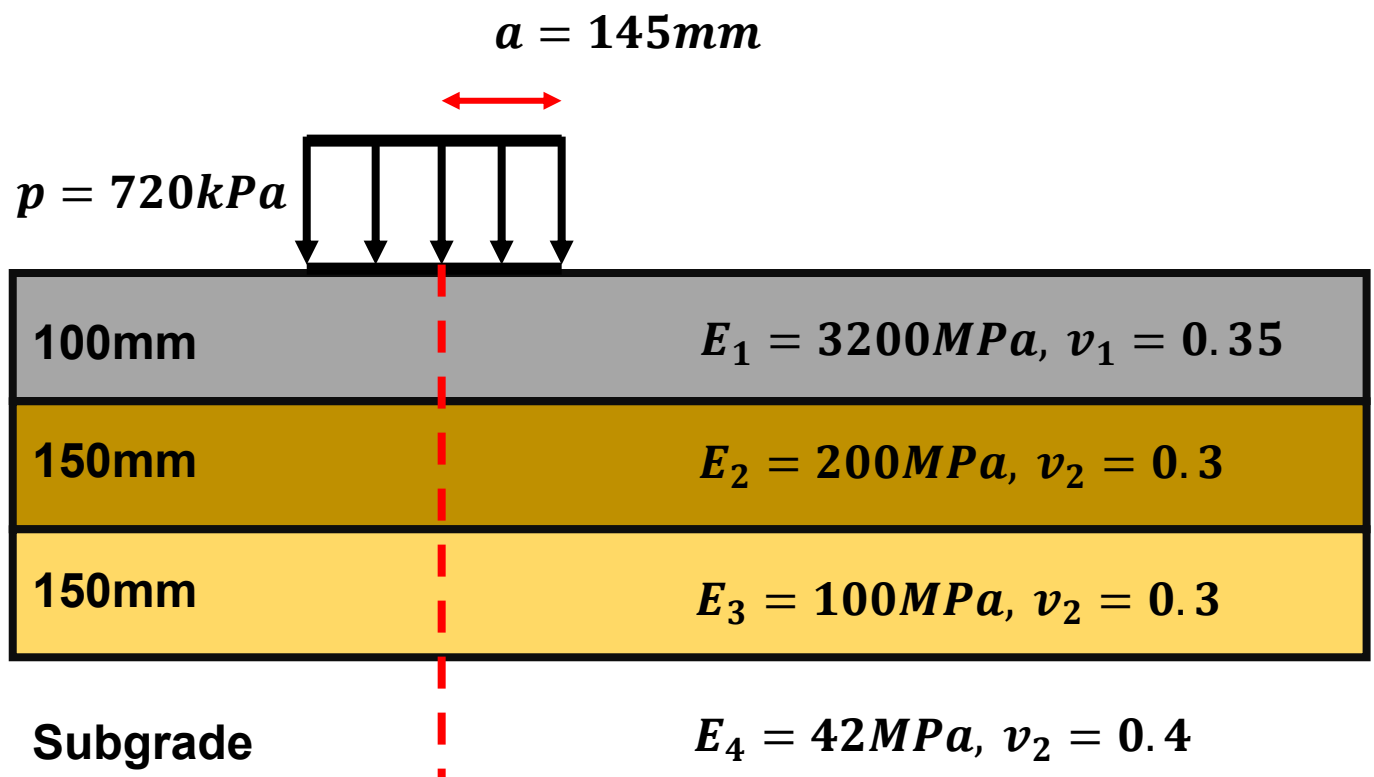


Figure 1. Pavement structure question 3